

Suggested Questions: 2vi, 3c, 4c, 4f, 5 and 6 and.

Qu. 1. Define what is meant by a vertical section and a level curve of a function of two variables x and y .

Let $z = f(x, y)$ where $f(x, y) = \frac{x^2 + y^2}{x}$. Sketch the vertical sections and level curves of f associated with the constants 0, 1 and 2.

Qu. 2. Assuming that the following equations define z as a function of the other variables occurring, use implicit differentiation to find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ where

$$(i) \quad x^2 - y^2 + z^2 = 1 \quad (ii) \quad z^2 - 4xz + y^2 = 4 \quad (iii) \quad xy + xz + yz = 3$$

$$(iv) \quad x^2 + z^2 = 4yz \quad (v) \quad xyz = x + y + z \quad (vi) \quad x^3 z^2 - z^2 y^2 = \sin x.$$

Qu. 3. Show that $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$ in the following cases by explicitly calculating all mixed second order derivatives of

- a) $f(x, y) = x^2 y^3$,
- b) $f(x, y) = e^{xy} \cos x$,
- c) $f(x, y, z) = x e^{-z} - 3xy$.

Qu. 4. Assuming that the functions are sufficiently smooth, use the chain rule to find the following:

- a) $\frac{dz}{dt}$ where $z = f(x, y)$, $x = g(t)$ and $y = h(t)$;
- b) $\frac{dz}{dt}$ where $z = f(x, y, t)$, $x = g(t)$ and $y = h(t)$;
- c) $\frac{\partial}{\partial x} f(2x + y, y e^x)$ in terms of the partial derivatives of f ;
- d) $\frac{\partial u}{\partial s}$ and $\frac{\partial u}{\partial t}$ when $u = \sqrt{x^2 + y}$, $x = e^{st}$ and $y = s + \sin t$;
- e) $\frac{\partial w}{\partial s}$ and if $w = f(x, y, z)$, where $x = g(s, t)$, $y = h(s, t)$ and $z = k(s)$;
- f) $\frac{\partial u}{\partial s}$ and $\frac{\partial u}{\partial t}$ when $u = \sqrt{x^2 + y^2}$, $x = e^{st}$ and $y = s + \sin t$.

Qu. 5. Find

$$\frac{\partial u}{\partial x}, \quad \frac{\partial u}{\partial y}, \quad \frac{\partial^2 u}{\partial x^2}, \quad \frac{\partial^2 u}{\partial y^2}, \quad \text{and} \quad \frac{\partial^2 u}{\partial x \partial y},$$

where

- a) $u = x^y$
- b) $u = \ln(x + y^2)$

(Hint: it might be a good idea to check the procedure by calculating the mixed derivatives in two ways, i.e. by differentiating with respect to x first and y second and the other way round.)

Qu. 6. In the 2nd-order partial differential equation

$$\frac{\partial^2 z}{\partial x^2} - y \frac{\partial^2 z}{\partial y^2} = \frac{1}{2} \frac{\partial z}{\partial y},$$

change the variables $(x, y) \mapsto (u, v)$, $u = x - 2\sqrt{y}$, $v = x + 2\sqrt{y}$ (implying obviously that $y > 0$).

Qu. 7. Find how the equation

$$\frac{\partial^2 w}{\partial x \partial y} + w = 0$$

will be transformed if we look for a particular solution in the form $w = f(u)$, where $u = (x - x_0)(y - y_0)$ (x_0 and y_0 are constants). (In other words, what form will this equation take if we write it down in terms of f and u only?)